# The University of Azad Jammu and Kashmir, Muzaffarabad

## Department of Software Engineering




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# Title: SDA Lab 02 - Domain Modeling Library System:

## TASK 1: 10 Candidate Conceptual Classes

1. Author: Person who  writes books.
2. Book: The conceptual book with title and ISBN.
3. Category: Book category like CS, History
4. BookCopy: Physical copy of book with barcode and status.
5. Member: Library member who borrows books.
6. Membership:  Membership plan start and expiry dates.
7. Librarian: Staff who manages loans and reservations.
8. Loan: Transactions representing borrowing of a book copy by a member.
9. Reservation: Request to hold a book that is currently unavailable.
10. Fine: Late return will be fined.

## TASK 2: For 3 Important Classes - Attributes + Methods

### 1. Book

- Attributes: isbn: String, title: String, publicationYear: int, language: String

- Methods: getTitle(): String, getBookDetails(): String

### 2. Member

- Attributes: memberId: String, name: String, email: String, phone: String

- Methods: register(): void, updateProfile(): void

### 3. Loan

- Attributes: `loanId: String, loanDate: Date, dueDate: Date, returnDate: Date`

- Methods: calculateFine(): double, isOverdue(): boolean

## TASK 3: Abstraction Decisions

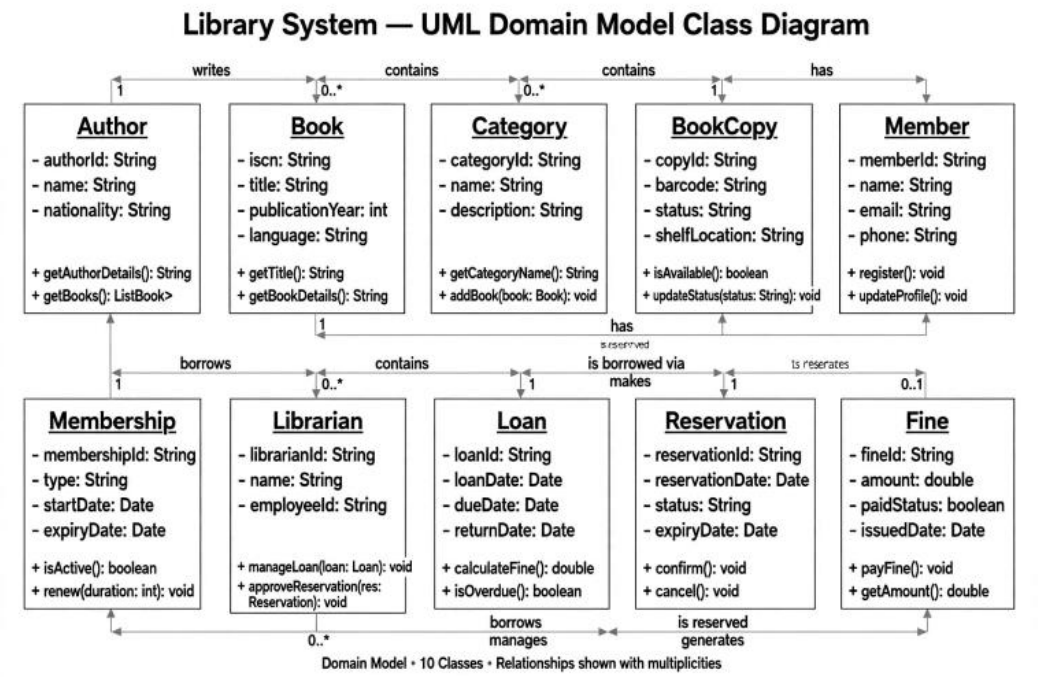
We ignored low-level details like book page count, font, paper quality, library furniture, and building. We also ignored member's age, gender, home address and fine payment method (JazzCash, cash). Our focus is only on domain concepts needed for borrowing/returning flow, not on UI or database implementation. BookCopy is kept separate from Book to handle multiple physical copies - this is a key abstraction.

## TASK 4: One "Invented" Software Object

Name: LoanManager / LibraryController / FineCalculator

This object does NOT exist in a real physical library. In real library, this does not exist. But in software we need an invented controller that checks "Is membership active? Is book available? Has member already borrowed 5 books?". It also automatically creates a Fine when Loan is overdue. We invent it to handle business logic that no single real-world person handles alone.

# Class Diagram:




Pitfalls Avoided:

1. We did NOT make Name, Date, Email as classes (they are attributes).

2. We did NOT make a God Class like `LibrarySystem`.

3. We kept identity: Two BookCopy with same title are different because copyId is different.

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